


# Pre-algebra 2e.

# Prealgebra 2e

SENIOR CONTRIBUTING AUTHORS
**Lynn Marecek, Santa Ana College**
**MaryAnne Anthony-Smith, Formerly of Santa Ana College**
**Andrea Honeycutt Mathis, Northeast Mississippi Community College**



**OpenStax**
Rice University
6100 Main Street MS-375
Houston, Texas 77005

To learn more about OpenStax, visit https://openstax.org.
Individual print copies and bulk orders can be purchased through our website.

**©2026 Rice University.** Textbook content produced by OpenStax is licensed under a Creative Commons Attribution Non-Commercial ShareAlike 4.0 International License (CC BY-NC-SA 4.0). Under this license, any user of this textbook or the textbook contents herein can share, remix, and build upon the content for noncommercial purposes only. Any adaptations must be shared under the same type of license. In any case of sharing the original or adapted material, whether in whole or in part, the user must provide proper attribution as follows:

- If you noncommercially redistribute this textbook in a digital format (including but not limited to PDF and HTML), then you must retain on every page the following attribution:
  "Access for free at openstax.org."
- If you noncommercially redistribute this textbook in a print format, then you must include on every physical page the following attribution:
  "Access for free at openstax.org."
- If you noncommercially redistribute part of this textbook, then you must retain in every digital format page view (including but not limited to PDF and HTML) and on every physical printed page the following attribution:
  "Access for free at openstax.org."
- If you use this textbook as a bibliographic reference, please include https://openstax.org/details/books/prealgebra-2e in your citation.

For questions regarding this licensing, please contact support@openstax.org.

**Trademarks**
The OpenStax name, OpenStax logo, OpenStax book covers, OpenStax CNX name, OpenStax CNX logo, OpenStax Tutor name, Openstax Tutor logo, Connexions name, Connexions logo, Rice University name, and Rice University logo are not subject to the license and may not be reproduced without the prior and express written consent of Rice University.

Kendall Hunt and the Kendall Hunt Logo are trademarks of Kendall Hunt. The Kendall Hunt mark is registered in the United States, Canada, and the European Union. These trademarks may not be used without the prior and express written consent of Kendall Hunt.

| | |
|---|---|
| **COLOR PAPERBACK BOOK ISBN-13** | **978-0-9986257-9-9** |
| **B&W PAPERBACK BOOK ISBN-13** | **978-1-975076-43-6** |
| **DIGITAL VERSION ISBN-13** | **978-1-951693-19-0** |
| **ORIGINAL PUBLICATION YEAR** | **2020** |

2 3 4 5 6 7 8 9 10 RS 26 20

## OpenStax

OpenStax provides free, peer-reviewed, openly licensed textbooks for introductory college and Advanced Placement® courses and low-cost, personalized courseware that helps students learn. A nonprofit ed tech initiative based at Rice University, we're committed to helping students access the tools they need to complete their courses and meet their educational goals.

## Rice University

OpenStax is an initiative of Rice University. As a leading research university with a distinctive commitment to undergraduate education, Rice University aspires to path-breaking research, unsurpassed teaching, and contributions to the betterment of our world. It seeks to fulfill this mission by cultivating a diverse community of learning and discovery that produces leaders across the spectrum of human endeavor.




## Philanthropic Support

OpenStax is grateful for the generous philanthropic partners who advance our mission to improve educational access and learning for everyone. To see the impact of our supporter community and our most updated list of partners, please visit openstax.org/foundation.

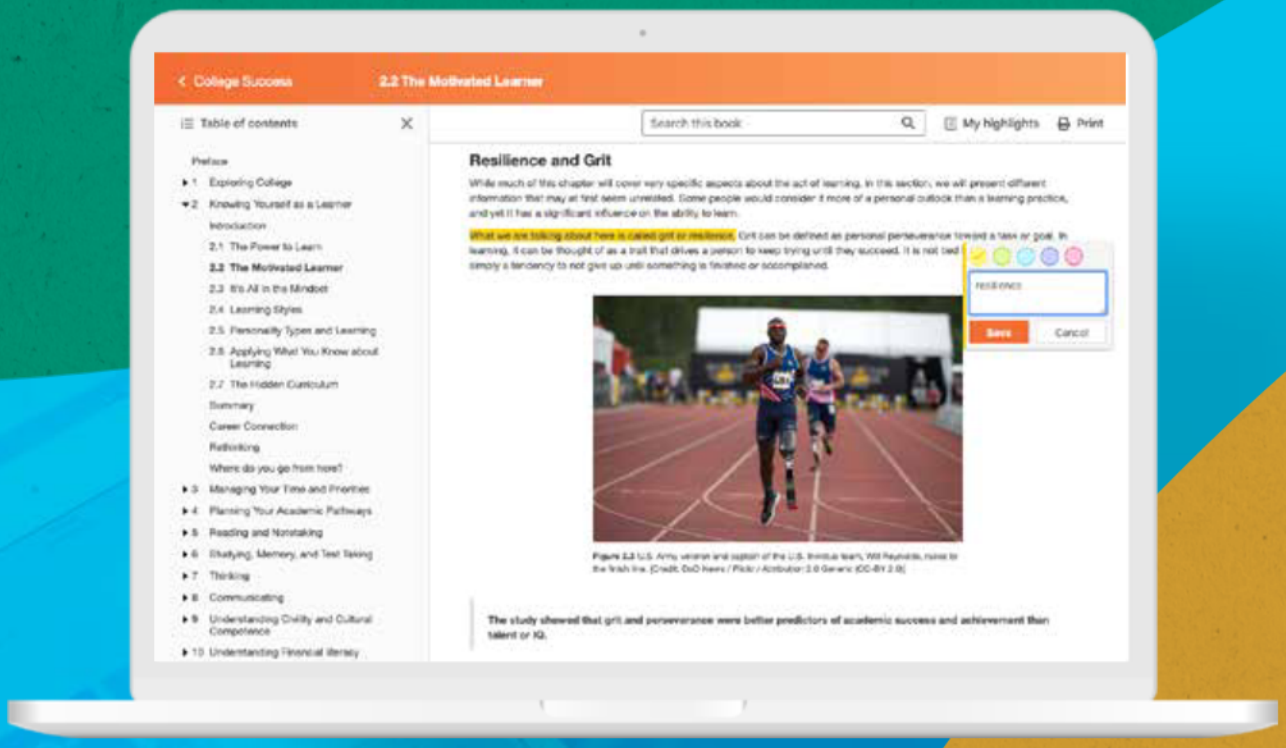
Arnold Ventures
Chan Zuckerberg Initiative
Chegg, Inc.
Arthur and Carlyse Ciocca Charitable Foundation
Digital Promise
Ann and John Doerr
Bill & Melinda Gates Foundation
Girard Foundation
Google Inc.
The William and Flora Hewlett Foundation
The Hewlett-Packard Company
Intel Inc.
Rusty and John Jaggers
The Calvin K. Kazanjian Economics Foundation
Charles Koch Foundation
Leon Lowenstein Foundation, Inc.
The Maxfield Foundation
Burt and Deedee McMurtry
Michelson 20MM Foundation
National Science Foundation
The Open Society Foundations
Jumee Yhu and David E. Park III
Brian D. Patterson USA-International Foundation
The Bill and Stephanie Sick Fund
Steven L. Smith & Diana T. Go
Stand Together
Robin and Sandy Stuart Foundation
The Stuart Family Foundation
Tammy and Guillermo Treviño
Valhalla Charitable Foundation
White Star Education Foundation
Schmidt Futures
William Marsh Rice University

openstax™

# Study where you want, what you want, when you want.

When you access your book in our web view, you can use our new online highlighting and note-taking features to create your own study guides.

Our books are free and flexible, forever.
Get started at openstax.org/details/books/prealgebra-2e




Access. The future of education.
openstax.org

RICE

# Contents


Preface 1

1 Whole Numbers 7

Introduction 7
**1.1** Introduction to Whole Numbers 7
**1.2** Add Whole Numbers 22
**1.3** Subtract Whole Numbers 38
**1.4** Multiply Whole Numbers 51
**1.5** Divide Whole Numbers 68
Chapter Review 85
Exercises 87

2 The Language of Algebra 95

Introduction to the Language of Algebra 95
**2.1** Use the Language of Algebra 95
**2.2** Evaluate, Simplify, and Translate Expressions 113
**2.3** Solving Equations Using the Subtraction and Addition Properties of Equality 126
**2.4** Find Multiples and Factors 139
**2.5** Prime Factorization and the Least Common Multiple 152
Chapter Review 162
Exercises 165

3 Integers 171

Introduction to Integers 171
**3.1** Introduction to Integers 171
**3.2** Add Integers 186
**3.3** Subtract Integers 201
**3.4** Multiply and Divide Integers 219
**3.5** Solve Equations Using Integers; The Division Property of Equality 232
Chapter Review 244
Exercises 246

4 Fractions 253

Introduction to Fractions 253
**4.1** Visualize Fractions 253
**4.2** Multiply and Divide Fractions 274
**4.3** Multiply and Divide Mixed Numbers and Complex Fractions 291
**4.4** Add and Subtract Fractions with Common Denominators 302
**4.5** Add and Subtract Fractions with Different Denominators 311
**4.6** Add and Subtract Mixed Numbers 329
**4.7** Solve Equations with Fractions 344
Chapter Review 359
Exercises 362

5 Decimals 367

Introduction to Decimals 367
**5.1** Decimals 367
**5.2** Decimal Operations 383
**5.3** Decimals and Fractions 401
**5.4** Solve Equations with Decimals 413
**5.5** Averages and Probability 421
**5.6** Ratios and Rate 433
**5.7** Simplify and Use Square Roots 444
Chapter Review 457
Exercises 460

6 Percents 467

Introduction to Percents 467
**6.1** Understand Percent 467
**6.2** Solve General Applications of Percent 482
**6.3** Solve Sales Tax, Commission, and Discount Applications 494
**6.4** Solve Simple Interest Applications 507
**6.5** Solve Proportions and their Applications 516
Chapter Review 532
Exercises 533

7 The Properties of Real Numbers 539

Introduction to the Properties of Real Numbers 539
**7.1** Rational and Irrational Numbers 539
**7.2** Commutative and Associative Properties 546
**7.3** Distributive Property 556
**7.4** Properties of Identity, Inverses, and Zero 567
**7.5** Systems of Measurement 578
Chapter Review 595
Exercises 596

8 Solving Linear Equations 601

Introduction to Solving Linear Equations 601
**8.1** Solve Equations Using the Subtraction and Addition Properties of Equality 601
**8.2** Solve Equations Using the Division and Multiplication Properties of Equality 614
**8.3** Solve Equations with Variables and Constants on Both Sides 621
**8.4** Solve Equations with Fraction or Decimal Coefficients 637
Chapter Review 647
Exercises 648

9 Math Models and Geometry 651

Introduction 651
**9.1** Use a Problem Solving Strategy 651
**9.2** Solve Money Applications 666
**9.3** Use Properties of Angles, Triangles, and the Pythagorean Theorem 679

**9.4** Use Properties of Rectangles, Triangles, and Trapezoids 698
**9.5** Solve Geometry Applications: Circles and Irregular Figures 724
**9.6** Solve Geometry Applications: Volume and Surface Area 736
**9.7** Solve a Formula for a Specific Variable 755
Chapter Review 766
Exercises 769

10 **Polynomials 779**

Introduction to Polynomials 779
**10.1** Add and Subtract Polynomials 779
**10.2** Use Multiplication Properties of Exponents 788
**10.3** Multiply Polynomials 801
**10.4** Divide Monomials 814
**10.5** Integer Exponents and Scientific Notation 830
**10.6** Introduction to Factoring Polynomials 845
Chapter Review 857
Exercises 859

11 **Graphs 865**

Graphs 865
**11.1** Use the Rectangular Coordinate System 865
**11.2** Graphing Linear Equations 886
**11.3** Graphing with Intercepts 904
**11.4** Understand Slope of a Line 919
Chapter Review 946
Exercises 947

A **Cumulative Review 955**

B **Powers and Roots Tables 961**

C **Geometric Formulas 967**

**Answer Key 969**

**Index 1063**

# Preface

Welcome to *Prealgebra 2e*, an OpenStax resource. This textbook was written to increase student access to high-quality learning materials, maintaining highest standards of academic rigor at little to no cost.

## About OpenStax

OpenStax is part of Rice University, which is a 501(c)(3) nonprofit charitable corporation. Our mission is to make an amazing education accessible for all. Through our partnerships with philanthropic organizations and our alliance with other educational resource companies, we're breaking down the most common barriers to learning. Because we believe that everyone should and can have access to knowledge.

### About OpenStax Resources

#### Customization

*Prealgebra 2e* is licensed under a Creative Commons Attribution-NonCommercial-ShareAlike 4.0 (CC BY NC-SA) license, which means that you can non-commercially distribute, remix, and build upon the content, as long as you provide attribution to OpenStax and its content contributors, and distribute all derivatives under the same license.

Because our books are openly licensed, you are free to use the entire book or pick and choose the sections that are most relevant to the needs of your course. Feel free to remix the content by assigning your students certain chapters and sections in your syllabus, in the order that you prefer. You can even provide a direct link in your syllabus to the sections in the web view of your book.

Instructors also have the option of creating a customized version of their OpenStax book. The custom version can be made available to students in low-cost print or digital form through their campus bookstore. Visit your book page on openstax.org for more information.

#### Art attribution in *Prealgebra 2e*

In *Prealgebra 2e*, most art contains attribution to its title, creator or rights holder, host platform, and license within the caption. Some art has been provided through permissions and should only be used with the attribution or limitations provided in the credit. For any art that is openly licensed, non-commercial users or organizations may reuse the art as long as they provide the same attribution to its original source. (Commercial entities should contact OpenStax to discuss reuse rights and permissions.) To maximize readability and content flow, some art does not include attribution in the text. If you reuse art from this text that does not have attribution provided, use the following attribution: Copyright Rice University, OpenStax, under CC BY-NC-SA 4.0 license.

#### Errata

All OpenStax textbooks undergo a rigorous review process. However, like any professional-grade textbook, errors sometimes occur. Since our books are web based, we can make updates periodically when deemed pedagogically necessary. If you have a correction to suggest, submit it through the link on your book page on openstax.org. Subject matter experts review all errata suggestions. OpenStax is committed to remaining transparent about all updates, so you will also find a list of past errata changes on your book page on openstax.org.

#### Format

You can access this textbook for free in web view or PDF through openstax.org, and for a low cost in print.

### About *Prealgebra 2e*

*Prealgebra 2e* is designed to meet scope and sequence requirements for a one-semester prealgebra course. The text introduces the fundamental concepts of algebra while addressing the needs of students with diverse backgrounds and learning styles. Each topic builds upon previously developed material to demonstrate the cohesiveness and structure of mathematics.

Students who are taking basic mathematics and prealgebra classes in college present a unique set of challenges. Many students in these classes have been unsuccessful in their prior math classes. They may think they know some math, but their core knowledge is full of holes. Furthermore, these students need to learn much more than the course content. They need to learn study skills, time management, and how to deal with math anxiety. Some students lack basic reading and arithmetic skills. The organization of *Prealgebra* makes it easy to adapt the book to suit a variety of course syllabi.

### Coverage and Scope

*Prealgebra 2e* takes a student-support approach in its presentation of the content. The beginning, in particular, is presented as a sequence of small steps so that students gain confidence in their ability to succeed in the course. The order of topics was carefully planned to emphasize the logical progression throughout the course and to facilitate a

thorough understanding of each concept. As new ideas are presented, they are explicitly related to previous topics.

- **Chapter 1: Whole Numbers**
  Each of the four basic operations with whole numbers—addition, subtraction, multiplication, and division—is modeled and explained. As each operation is covered, discussions of algebraic notation and operation signs, translation of algebraic expressions into word phrases, and the use of the operation in applications are included.
- **Chapter 2: The Language of Algebra**
  Mathematical vocabulary as it applies to the whole numbers is presented. The use of variables, which distinguishes algebra from arithmetic, is introduced early in the chapter, and the development of and practice with arithmetic concepts use variables as well as numeric expressions. In addition, the difference between expressions and equations is discussed, word problems are introduced, and the process for solving one-step equations is modeled.
- **Chapter 3: Integers**
  While introducing the basic operations with negative numbers, students continue to practice simplifying, evaluating, and translating algebraic expressions. The Division Property of Equality is introduced and used to solve one-step equations.
- **Chapter 4: Fractions**
  Fraction circles and bars are used to help make fractions real and to develop operations on them. Students continue simplifying and evaluating algebraic expressions with fractions, and learn to use the Multiplication Property of Equality to solve equations involving fractions.
- **Chapter 5: Decimals**
  Basic operations with decimals are presented, as well as methods for converting fractions to decimals and vice versa. Averages and probability, unit rates and unit prices, and square roots are included to provide opportunities to use and round decimals.
- **Chapter 6: Percents**
  Conversions among percents, fractions, and decimals are explored. Applications of percent include calculating sales tax, commission, and simple interest. Proportions and solving percent equations as proportions are addressed as well.
- **Chapter 7: The Properties of Real Numbers**
  The properties of real numbers are introduced and applied as a culmination of the work done thus far, and to prepare students for the upcoming chapters on equations, polynomials, and graphing.
- **Chapter 8: Solving Linear Equations**
  A gradual build-up to solving multi-step equations is presented. Problems involve solving equations with constants on both sides, variables on both sides, variables and constants on both sides, and fraction and decimal coefficients.
- **Chapter 9: Math Models and Geometry**
  The chapter begins with opportunities to solve “traditional” number, coin, and mixture problems. Geometry sections cover the properties of triangles, rectangles, trapezoids, circles, irregular figures, the Pythagorean Theorem, and volumes and surface areas of solids. Distance-rate-time problems and formulas are included as well.
- **Chapter 10: Polynomials**
  Adding and subtracting polynomials is presented as an extension of prior work on combining like terms. Integer exponents are defined and then applied to scientific notation. The chapter concludes with a brief introduction to factoring polynomials.
- **Chapter 11: Graphs**
  This chapter is placed last so that all of the algebra with one variable is completed before working with linear equations in two variables. Examples progress from plotting points to graphing lines by making a table of solutions to an equation. Properties of vertical and horizontal lines and intercepts are included. Graphing linear equations at the end of the course gives students a good opportunity to review evaluating expressions and solving equations.

All chapters are broken down into multiple sections, the titles of which can be viewed in the Table of Contents.

## Changes to the Second Edition

The *Prealgebra 2e* revision focused on mathematical clarity and accuracy. Every Example, Try-It, Section Exercise, Review Exercise, and Practice Test item was reviewed by multiple faculty experts, and then verified by authors. This intensive effort resulted in hundreds of changes to the text, problem language, answers, instructor solutions, and graphics.

However, OpenStax and our authors are aware of the difficulties posed by shifting problem and exercise numbers when

textbooks are revised. In an effort to make the transition to the 2nd edition as seamless as possible, we have minimized any shifting of exercise numbers. For example, instead of deleting or adding problems where necessary, we replaced problems in order to keep the numbering intact. As a result, in nearly all chapters, there will be no shifting of exercise numbers; in the chapters where shifting does occur, it will be minor. Faculty and course coordinators should be able to use the new edition in a straightforward manner.

Also, to increase convenience, answers to the Be Prepared Exercises will now appear in the regular solutions manuals, rather than as a separate resource.

A detailed transition guide is available as an instructor resource at openstax.org.

## Pedagogical Foundation and Features

### *Learning Objectives*

Each chapter is divided into multiple sections (or modules), each of which is organized around a set of learning objectives. The learning objectives are listed explicitly at the beginning of each section and are the focal point of every instructional element.

### *Narrative text*

Narrative text is used to introduce key concepts, terms, and definitions, to provide real-world context, and to provide transitions between topics and examples. An informal voice was used to make the content accessible to students.

Throughout this book, we rely on a few basic conventions to highlight the most important ideas:

- Key terms are boldfaced, typically when first introduced and/or when formally defined.
- Key concepts and definitions are called out in a blue box for easy reference.

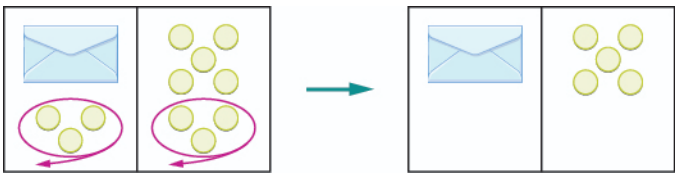
### *Examples*

Each learning objective is supported by one or more worked examples, which demonstrate the problem-solving approaches that students must master. Typically, we include multiple Examples for each learning objective in order to model different approaches to the same type of problem, or to introduce similar problems of increasing complexity.

All Examples follow a simple two- or three-part format. First, we pose a problem or question. Next, we demonstrate the Solution, spelling out the steps along the way. Finally (for select Examples), we show students how to check the solution. Most examples are written in a two-column format, with explanation on the left and math on the right to mimic the way that instructors “talk through” examples as they write on the board in class.

### *Figures*

*Prealgebra 2e* contains many figures and illustrations. Art throughout the text adheres to a clear, understated style, drawing the eye to the most important information in each figure while minimizing visual distractions.



### *Supporting Features*

Four small but important features serve to support Examples:

#### Be Prepared!

Each section, beginning with Section 1.2, starts with a few “Be Prepared!” exercises so that students can determine if they have mastered the prerequisite skills for the section. Reference is made to specific Examples from previous sections so students who need further review can easily find explanations. Answers to these exercises can be found in the supplemental resources that accompany this title.

#### How To



A “How To” is a list of steps necessary to solve a certain type of problem. A "How To" typically precedes an Example.

#### Try It



A “Try It” exercise immediately follows an Example, providing the student with an immediate opportunity to solve a similar problem. In the PDF and the Web View version of the text, answers to the Try It exercises are located in the Answer Key.

**Media**



The “Media” icon appears at the conclusion of each section, just prior to the Section Exercises. This icon marks a list of links to online video tutorials that reinforce the concepts and skills introduced in the section.

Disclaimer: While we have selected tutorials that closely align to our learning objectives, we did not produce these tutorials, nor were they specifically produced or tailored to accompany *Prealgebra 2e*.

### *Section Exercises*

Each section of every chapter concludes with a well-rounded set of exercises that can be assigned as homework or used selectively for guided practice. Exercise sets are named Practice Makes Perfect to encourage completion of homework assignments.

- Exercises correlate to the learning objectives. This facilitates assignment of personalized study plans based on individual student needs.
- Exercises are carefully sequenced to promote building of skills.
- Values for constants and coefficients were chosen to practice and reinforce arithmetic facts.
- Even and odd-numbered exercises are paired.
- Exercises parallel and extend the text examples and use the same instructions as the examples to help students easily recognize the connection.
- Applications are drawn from many everyday experiences, as well as those traditionally found in college math texts.
- **Everyday Math** highlights practical situations using the math concepts from that particular section.
- **Writing Exercises** are included in every Exercise Set to encourage conceptual understanding, critical thinking, and literacy.

### *Chapter Review Features*

The end of each chapter includes a review of the most important takeaways, as well as additional practice problems that students can use to prepare for exams.

- **Key Terms** provides a formal definition for each bold-faced term in the chapter.
- **Key Concepts** summarizes the most important ideas introduced in each section, linking back to the relevant Example(s) in case students need to review.
- **Chapter Review Exercises** includes practice problems that recall the most important concepts from each section.
- **Practice Test** includes additional problems assessing the most important learning objectives from the chapter.
- **Answer Key** includes the answers to all Try It exercises and every other exercise from the Section Exercises, Chapter Review Exercises, and Practice Test.

### Answers to Questions in the Book

All answers to Try It questions are provided in the Answer Key. Answers to Examples are provided directly below the question. Students can find odd-numbered answers to Chapter Review Exercises, Practice Test, and Section Exercises in the Answer Key. Answers to all odd and even-numbered questions are provided only to instructors in the Instructor Answer Guide via the Instructor Resources page.

## Additional Resources

### Student and Instructor Resources

We’ve compiled additional resources for both students and instructors, including Getting Started Guides, manipulative mathematics worksheets, Links to Literacy assignments, and an answer key. Instructor resources require a verified instructor account, which can be requested on your openstax.org log-in. Take advantage of these resources to supplement your OpenStax book.

### Partner Resources

OpenStax Partners are our allies in the mission to make high-quality learning materials affordable and accessible to students and instructors everywhere. Their tools integrate seamlessly with our OpenStax titles at a low cost. To access the partner resources for your text, visit your book page on openstax.org.

# About the Authors

## Senior Contributing Authors

Lynn Marecek and MaryAnne Anthony-Smith taught mathematics at Santa Ana College for many years and have worked together on several projects aimed at improving student learning in developmental math courses. They are the authors of *Strategies for Success: Study Skills for the College Math Student,* published by Pearson HigherEd.

**Lynn Marecek, Santa Ana College**

**MaryAnne Anthony-Smith, Santa Ana College**

**Andrea Honeycutt Mathis, Northeast Mississippi Community College**

## Reviewers

Tony Ayers, Collin College Preston Ridge Campus
David Behrman, Somerset Community College
Brandie Biddy, Cecil College
Bryan Blount, Kentucky Wesleyan College
Steven Boettcher, Estrella Mountain Community College
Kimberlyn Brooks, Cuyahoga Community College
Pamela Burleson, Lone Star College University Park
Tamara Carter, Texas A&M University
Phil Clark, Scottsdale Community College
Christina Cornejo, Erie Community College
Denise Cutler, Bay de Noc Community College
Richard Darnell, Eastern Wyoming College
Robert Diaz, Fullerton College
Karen Dillon, Thomas Nelson Community College
Valeree Falduto, Palm Beach State
Bryan Faulkner, Ferrum College
David French, Tidewater Community College
Stephanie Gable, Columbus State University
Heather Gallacher, Cleveland State University
Rachel Gross, Towson University
Dianne Hendrickson, Becker College
Linda Hunt, Shawnee State University
Betty Ivory, Cuyahoga Community College
Joanne Kendall, Lone Star College System
Kevin Kennedy, Athens Technical College
Stephanie Krehl, Mid-South Community College
Allyn Leon, Imperial Valley College
Gerald LePage, Bristol Community College
Laurie Lindstrom, Bay de Noc Community College
Jonathan Lopez, Niagara University
Yixia Lu, South Suburban College
Mikal McDowell, Cedar Valley College
Kim McHale, Columbia College of Missouri
Allen Miller, Northeast Lakeview College
Michelle Moravec, Baylor University TX/McLennan Community College
Jennifer Nohai-Seaman, Housatonic Community College
Rick Norwood, East Tennessee State University
Linda Padilla, Joliet Junior College
Kelly Proffitt, Patrick Henry Community College
Teresa Richards, Butte-Glenn Community College
Christian Roldan-Johnson, College of Lake County Community College
Patricia C. Rome, Delgado Community College, City Park Campus
Kegan Samuel, Naugatuck Valley Community College
Bruny Santiago, Tarrant College Southeast Campus
Sutandra Sarkar, Georgia State University
Richard Sgarlotti, Bay Mills Community College

Chuang Shao, Rose State College
Carla VanDeSande, Arizona State University
Shannon Vinson, Wake Technical Community College
Maryam Vulis, Norwalk Community College
Toby Wagner, Chemeketa Community College
Libby Watts, Tidewater Community College
Becky Wheelock, San Diego City College